

LAB 05

Network Reconnaissance & Service Enumeration

Security Operations Report

COURSE: **Managing Vulnerabilities**
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CLASSIFICATION: **Educational Use Only**

1. Introduction

This report documents a controlled network reconnaissance and service enumeration exercise performed in a virtualized lab environment. The purpose of this activity was to identify active hosts, enumerate exposed services, establish a baseline scan, perform a post-change scan, and analyze any differences. The exercise follows common blue-team and SOC analyst methodologies aligned with entry-level security operations practices.

Scope Notice: All activities were conducted strictly for educational purposes within an isolated host-only virtual network. No external systems were targeted or affected.

2. Lab Environment

Component	Details
Attacking Host	Kali Linux — Nmap 7.95
Target Network	192.168.56.0/24
Target Host (Focus)	192.168.56.106
Virtualization Platform	VirtualBox (Host-only network)
Target OS (Detected)	Linux — Debian-based

The network is a host-only virtual network, meaning no external internet exposure exists. This setup is commonly used for penetration testing and defensive security labs.

3. Host Discovery

Command executed:

```
nmap -sn 192.168.56.0/24 --reason
```

Four active hosts were identified on the network:

192.168.56.1	Network gateway
192.168.56.100	VirtualBox host
192.168.56.106	Target system (primary focus)
192.168.56.108	Localhost-related response

The target system at 192.168.56.106 responded with an ARP reply, confirming it is reachable and active on the local network.

4. Connectivity Verification (ICMP)

```
ping 192.168.56.106
```

Metric	Result	Status
Packet Loss	0%	✓ PASS
RTT Stability	Stable	✓ PASS
Incomplete IP Test (192.168.56.)	Name resolution error	✓ Expected

5. Baseline Service Scan

Full service version detection with OS fingerprinting and default scripts:

```
sudo nmap -sV -sC -O 192.168.56.106 -oA baseline_scan
```

Discovered Open Ports:

Port	Protocol	Service	Version	Status
22	TCP	SSH	OpenSSH 10.0	OPEN
80	TCP	HTTP	Apache 2.4.65	OPEN

Additional Findings:

- Apache default page accessible — indicates a stock/unconfigured web server
- SSH reachable with no access restrictions detected at scan level
- OS detection indicates Linux kernel versions 4.x – 5.x

6. Configuration Change Attempt

An attempt was made to restart the SSH service using sudo:

```
sudo systemctl restart ssh
```



Result: "slm is not in sudoers" — The current user does not have elevated privileges. The change was rejected by the system. This confirms that least-privilege controls are functioning correctly on the target host.

7. Post-Change Scan

A second scan was performed after the attempted configuration change:

```
nmap -sV -sC 192.168.56.106 -oA post_change_scan
```

The post-change scan returned identical service information as the baseline. Both port 22 (SSH) and port 80 (HTTP) remained open with the same version details, confirming no configuration changes occurred.

8. Scan Comparison

```
ndiff baseline_scan.xml post_change_scan.xml
```

Comparison Dimension	Baseline Scan	Post-Change Scan	Delta
Open Ports	22, 80	22, 80	None
SSH Version	OpenSSH 10.0	OpenSSH 10.0	None
HTTP Version	Apache 2.4.65	Apache 2.4.65	None
OS Detection	Linux 4.x–5.x	Linux 4.x–5.x	None
Timestamp	14 Feb 2026 17:44	14 Feb 2026 17:53	9 min

Result: Only timestamp differences were detected. No functional changes were observed.

9. Security Assessment

Service	Version	Risk Level	Finding	Recommendation
SSH	OpenSSH 10.0	MEDIUM	No access restrictions at scan level	Enforce key-based auth; restrict via firewall (iptables)
HTTP	Apache 2.4.65	HIGH	Default Apache page exposed — reveals server identity	Disable default page; configure virtual host

10. Conclusion

This lab exercise successfully validated the end-to-end reconnaissance and change-detection workflow commonly used in blue-team and SOC operations:

Step	Action	Outcome
1	Host Discovery	4 active hosts identified via ARP sweep
2	Connectivity Verification	Target confirmed reachable — 0% packet loss
3	Baseline Scan	SSH 10.0 + Apache 2.4.65 documented
4	Change Attempt	Blocked — sudoers policy enforced
5	Post-Change Scan	Identical results — system integrity confirmed
6	Comparison (ndiff)	No functional changes — only timestamp delta

KEY FINDING

Least-privilege access controls successfully prevented unauthorized service modifications, validating the target system's security posture.

Tools Used

Primary: nmap, ndiff | **Alternative:** RustScan